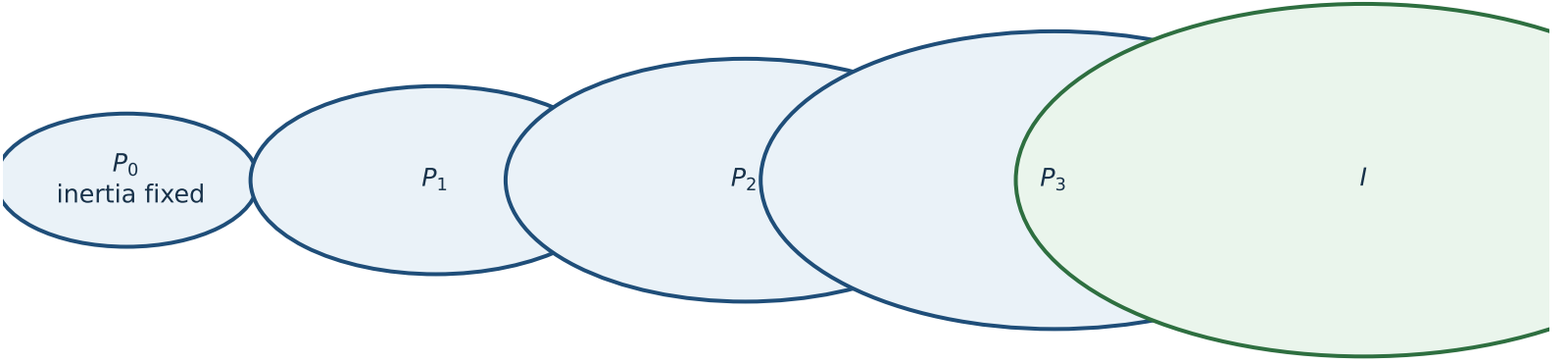
# Ramification is a nested projector resolution




$$a(V) = \sum_i \frac{|G_i|}{|G_0|} \operatorname{Tr}(I - P_i)$$

The Artin conductor is the weighted unresolved rank; the Swan conductor removes the tame level.